

[DC] Investigating student's learning processes in a Virtual Reality learning environment by analyzing their interaction behavior

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ABSTRACT

Educational researchers use physiological sensors and human observers to collect data to conduct quantitative studies in VR Learning Environments (VRLE). The data collected using physiological sensors are used to understand the affective state of the learners. Whereas, human observers are the most widely used method to collect data to understand the behavior of the learners in a VRLE. However, the data collection using human observers is labor intensive and may get biased due to cognitive, social, and communicative causes. Moreover, the data provided by human observers need to satisfy the inter-rater reliability test in order to become valid. There has been little or no work done to automatically collect the data related to the behavior of the learners in VRLE. To overcome the gaps mentioned above we have developed a data collection mechanism to automatically log all the interaction behavior of the learners happening in VRLE along with the time-stamp in real-time. The developed data collection mechanism can open up new opportunities for data collection when deployed in VRLEs. Also, the data collected using it can be useful to get more insights about the learners' behavior and in turn the learning processes happening in VRLE by applying AI and ML tools. This doctoral thesis aims at investigating the interaction behavior of the learners happening in VRLE and analyze the behavioral data to understand the behavioral pattern responsible for the higher learning gain. The analysis can also be further done for the early prediction of the learning outcomes based on the interaction behavior to provide scaffolds and adaptive feedback.

Index Terms: Interaction Behavior—Virtual Reality Learning Environment—pattern mining—modeling learners

1 INTRODUCTION AND RELATED WORK

Virtual Reality (VR) is a technology that immerses users in the virtual world and makes them interact with virtual objects to give them a more natural experience similar to that in the real world [9]. Thus VR technology has distinctive qualities including immersion, interaction, and imagination [6, 10]. The distinctive characteristics of VR have made the technology be used in various domains such as the Automotive industry, Military, Healthcare, Sports domain, etc including the education domain. In the education domain, VR is used to teach and learn content involving 1) concepts that can not be viewed by humans with the naked eye such as the strands of DNA [8], the function of the human artillery system [3], cell structure, etc, 2) places that can not be accessed by humans such as space, the primitive world, etc, and 3) the scenarios that are too hazardous and dangerous to experience in real life such as fire fighting [7], welding, oil refinery, etc.

Due to the above-mentioned qualities of VR, there is an exponential increase in the number of studies being conducted to analyze

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the use of VR in education in the last decade. Most of the empirical studies were conducted by collecting data from pre-test and post-test, self-reported questionnaires, interviews, and surveys [6], devices such as 1) physiological sensors [1], 2) eye trackers [9] 3) body trackers and 4) orientation of the head-mounted displays (HMD) and handheld controllers (HHC), and human observers [3]. Although studies are conducted to analyze the learning outcomes and learning processes based on the learners' psychological learning behavior (learning style), cognitive load, and user experience in VRLE [2], understanding the learning processes from the dynamic interaction behavior in VRLE is still at its infancy. This is mainly due to the non-existence of an efficient data collection mechanism that can log the learners' interaction behavior in VRLE. Hence, the goal of this doctoral thesis is to investigate the learning processes by collecting the interaction behavior of the learners. This can be accomplished by 1) creating a Virtual Reality Learning Environment (VRLE), 2) developing an interaction behavioral data (IBD) collection mechanism and deploying it in the VRLE, 3) Investigating the learning processes by analyzing the data collected, 4) providing suggestions to the VRLE developers to provide adaptive learning contents, feedback, and scaffolds to learners based on their behavior.

2 CURRENT PROGRESS

We have developed an IBD collection mechanism that is able to log all the interaction behavior of the learners in real time along with the time stamp [5] and deployed in the VRLE, MaroonVR [4].

2.1 Virtual Reality Learning Environment

We aim to collect IBD from the VRLE having conceptual learning content as it supports both rich interactions similar to procedural learning, and cognitive skill acquisition similar to factual learning. For this purpose, we choose MaroonVR, a VRLE developed by Johanna Pirker, and used to learn physics concepts through VR experiments such as Faraday's law, falling coil, Van de Graff generator, Huygens principle, etc. [4]. We adopted two scenes of MaroonVR that are used to perform 1) Faraday's law experiment and 2) the falling coil experiment to learn the notion of electromagnetic induction. Electromagnetic induction is a phenomenon in which the electromotive force (emf) is induced across a coil when a magnet is moved to and fro in and around the close proximity of the coil. The learners can interact by grabbing the magnet and dragging it through the coil and by varying the parameters such as the turns of the coil, the diameter of the coil, the resistance of the coil, magnetic strength, etc.

In addition to Faraday's law and falling coil experiment scenes, we included a new scene to boost embodiment learning. The learners engage with the VRLE while taking on the perspective of the magnet in the embodiment scene and the emf is induced due to their movement in and out of the coil. The learners can also visualize the induced emf plotted in real-time on the virtual graph present in all the scenes.

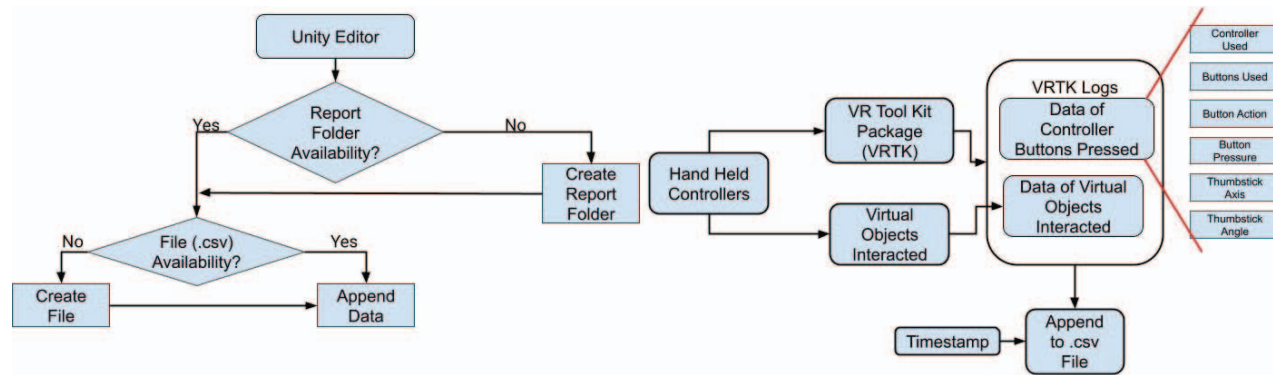


Figure 1: .csv File Creation (left) and Interaction Behavioral Data Collection Mechanism to log data in the created .csv file (right)

2.2 Interaction Behavioral Data Collection Mechanism

We developed the IBD collection mechanism as shown in Figure 1 using the Unity game engine [5]. The Virtual Reality Tool Kit (VRTK) package of Unity is able to collect the interactive actions performed on the buttons of HHCs such as controllers used, buttons used, button actions, button pressure, thumbstick axis, and thumbstick angle. The VRTK package, despite providing most of the information happening through the controllers, does not convey any information about the virtual objects interacted, and the time of occurrence of the interactive actions. Hence, we wrote a separate *c#* snippet in Unity to log the information related to the virtual object being interacted with when any of the HHC buttons are operated. A separate *c#* code is written in Unity to create a report folder and a report .csv file to which the virtual objects interacted and the buttons used during the interactions are appended. Also, the data logging mechanism is done in such a way that whenever a new entry is appended to the .csv file there is a separate column that records the time at which the entries are made. Thus the timestamp related to the interaction behavior of the learners also gets logged.

3 STUDY REPORT

To test the IBD collection mechanism and to understand how learners interact in the VRLE, we conducted a study with 14 undergraduate engineering students. The IBD collected is analyzed to understand the VR behavior of the learners such as the duration spent in VRLE, duration of interactions, duration of reading the instructions, frequency of parameter manipulation, frequency of interactions, and frequency of scene switching. The study result showed a normalized learning gain of 0.43. We also proved using a paired t-test that the significant rise in the post-test score is due to the VRLE intervention. In addition, we designed a linear regression model to predict the post-test score. R^2 value for the specified regression model was obtained as 0.487. So far, we have developed an IBD collection mechanism and deployed it in a VRLE to collect the IBD. It was found that the IBD collection mechanism is able to log all the interaction behavior of the learners along with the timestamp. The VRLE is also able to impart significant learning gain to the learners.

4 CONCLUSION

We conclude that the IBD collection mechanism is able to collect the IBD of the learners along with the timestamp. However, despite the fact that the study was effective in collecting IBD, it was carried out with a small sample size. Hence, further study with a larger sample size is needed. We also propose to analyze the data and perform process mining and behavioral pattern mining to identify the behavioral patterns responsible for high and low performance.

We also propose to use the results of the investigation to provide suggestions to the learners, instructors, and VRLE developers.

During the doctoral consortium, we expect suggestions and feedback related to our current progress and the proposed next steps. We specifically expect your recommendations on more behavioral data to model the VR learners and the ways to make the system more generalizable.

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